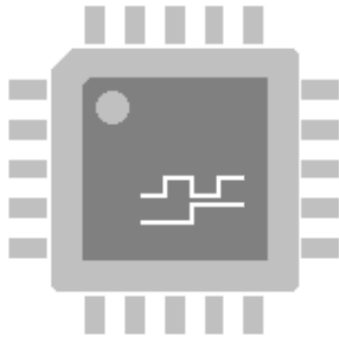


skSoC: a RISC-V Hazard3 based SoC skeleton



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Anton Duca



Motivation

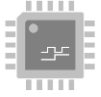
x86_64 dominates personal computers

ARM in smart phones

RISC-V in embedded systems

is expected to have a market of USD 1.41 billion in 2025, and 7 billion in 2030

has a free to implement ISA (Instruction Set Architecture); ARM and x86_64 don't



RISC-V specific

RISC-V has an extensible ISA

- RV32I is Its base integer unit for 32 bits
- RV64I is Its base integer unit for 64 bits

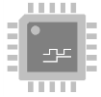
Extensions to the basis: IMACFDV...

- M for multiplication and division
- F and D for 32 and 64 bit floating point
- A for atomic read-modify-write operations
- C for compressed instructions
- V for vector



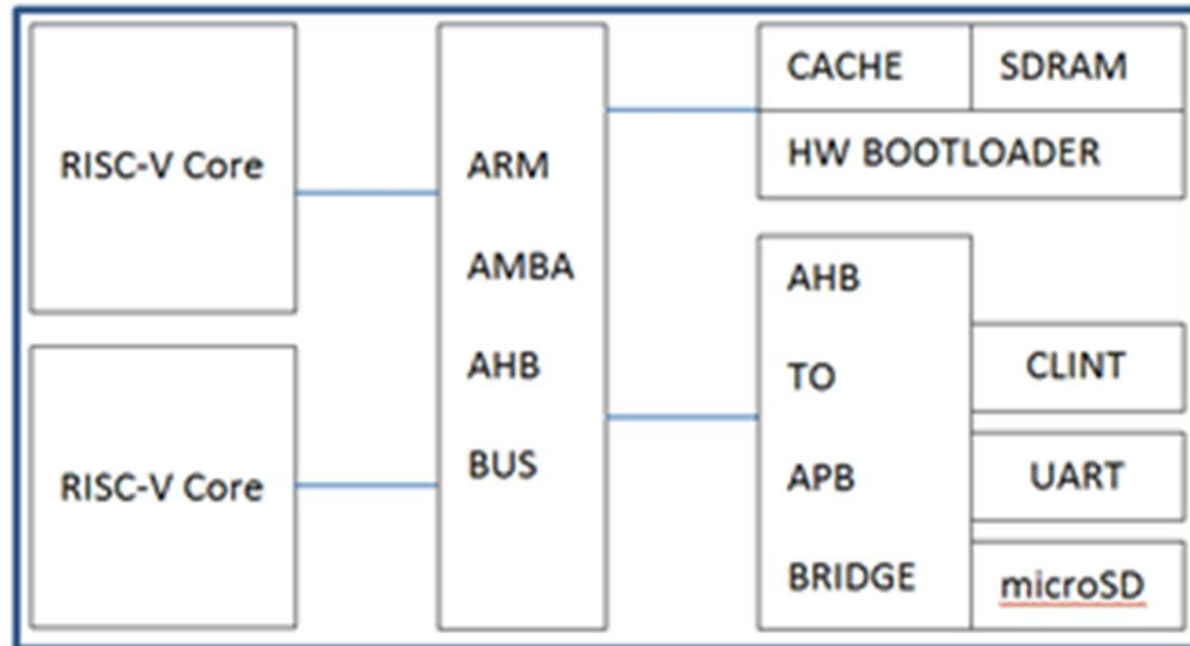
State of the art

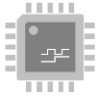
- commercial
 - SiFive, StarFive Tech, Microchip: RV64GC
 - Espressif Systems: RV32IM with bluetooth and wifi
- open source
 - Rocket Chip generator, BOOM in Chisel
 - CVA6
 - VexRISCV in SpinalHDL
 - Wally, NutShell single core with MMU
 - PicoRV32 non fully compatible with RISC-V spec
 - NOEL-V in VHDL
 - Hazard3 on RPi pico 2: the smallest: chosen



skSoC architecture

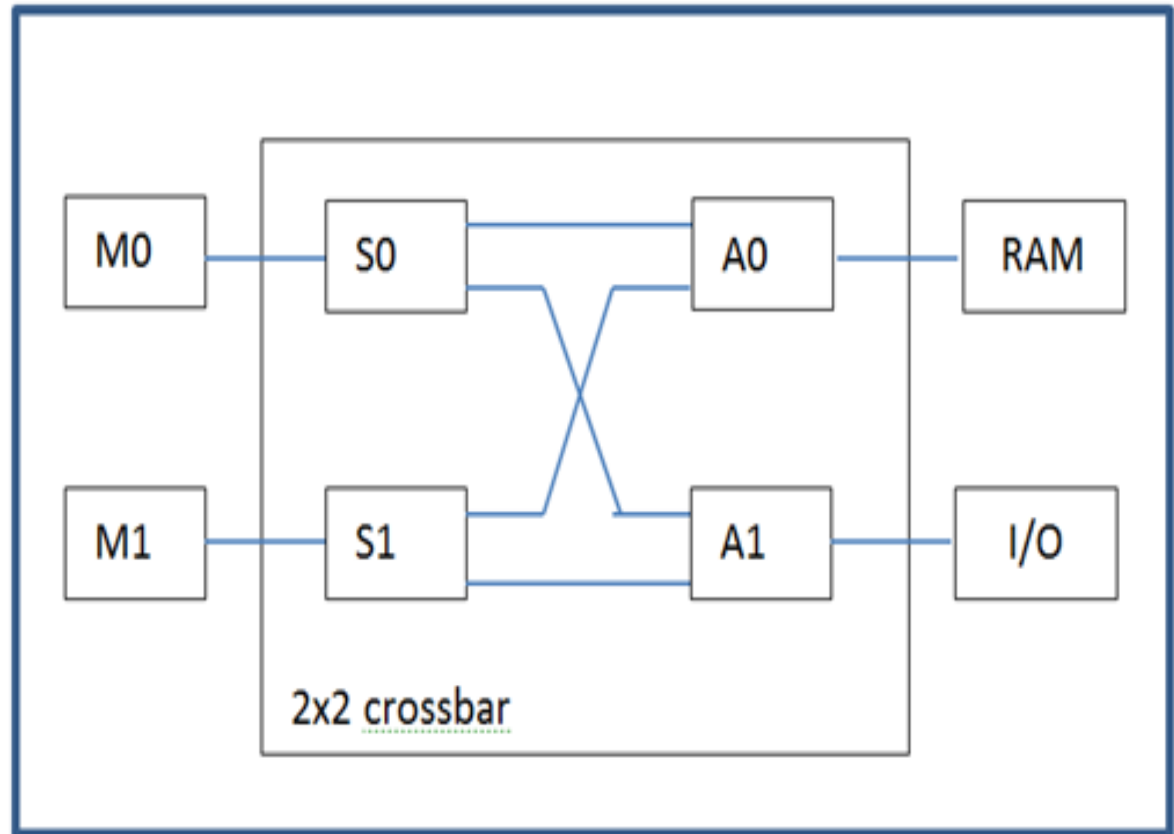
- one or two RISC-V RV32imac cores
- ARM AMBA AHB 5 bus
 - we implemented exclusive transfers for atomic instructions
- CLINT (common local interrupter)
- UART
- microSD
- cache, SDRAM
- HW bootloader





AMBA AHB bus 2x2 crossbar

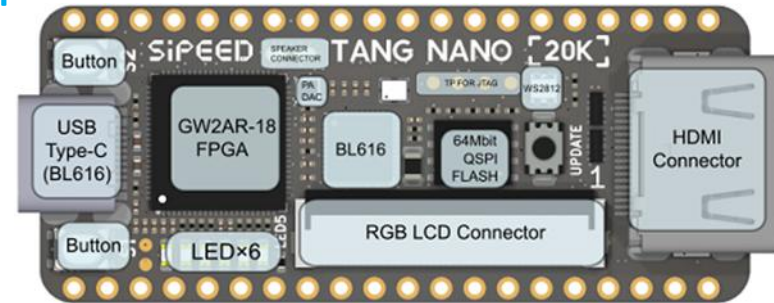
- M0=core0
- S0=splitter0
- A0=arbiter0
- RAM=slave0
- I/O=slave1

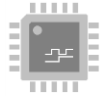




The FPGA board

- Tang nano 20k hardware board by Sipeed
- 39 USD, 27 MHz
- GW2AR-18 Gowin FPGA chip
- 20k LUTs, 15k flip-flops
- 828Kb of BlockSRAMs, 64Mb of SDRAM
- a permanent flash memory for FPGA config
- HDMI, microSD, 6 leds and 2 buttons





FPGA synthesis report

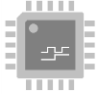
Resource	Usage	Utilization
Logic 1 core	6129 LUTs	33%
Logic 2 cores	11657 LUTs	64%
Registers	2525	16%
BSRAM	29	64%

- max 220MHz clock frequency



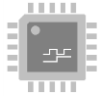
Operating System

- **hardware bootloader**; the microSD can be raw or FAT32 formatted
 - loads first 8MB of microSD into SDRAM, or
 - loads the initmem.bin from microSD
- we ported Nuttx RTOS
 - starting from the Nuttx flat RISC-V Qemu virtual board
- we wrote the microsd block device driver in Nuttx
- we enabled the Nuttx FAT32 file system
- for 256 KB Nuttx image, boots in less than 2s



Results (1)

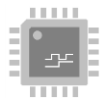
- id 0 bypasses the system bus (C program)
- id 1 about 128KB memory - all in FPGA RAM
- id 2/3 are for our implementation
 - with write-back/write-through cache
 - cache 32KB and 8MB RAM
- id 4 has no cache



Results dhrystone

id	Dhrystones/second
0	2638
1	664
2	503
3	376
4	67

- bus idle only 5% of test time



Results coremark

id	Iterations/second
0	3.8
1	1.20
2	1.02
3	0.82
4	-



Conclusion

- SoC dedicated to cheap FPGAs with a low number of LUTs
 - less than 6.2k LUTs for single core
 - performant,
 - with pipeline
 - being able to access 8MB of memory or more
- based on the RISC-V Hazard3 processor
- is written in Verilog
 - fast design, implementation, customization and testing